

Dive Buddy Alert Band

Development-partner brief / Presentation Rev G / CAD Rev F

Purpose

Get a paired diver's attention, receive their acknowledgement and, when a current reference fix is available, see where to look for them.

First customer and complete system

Proposed first market: operator-managed recreational boat dives. The operator owns and prepares the equipment. Divers wear two paired bands. A boat-associated acoustic reference supplies shared horizontal positions. An inductive dock charges the sealed modules between uses. Demand and operator workload remain unestablished.

First-product priorities

Pair identity, call, explicit human acknowledgement, clear missing-data states, adjustable retention and sealed charging lead the brief. Buddy bearing and relative depth depend on reference feasibility. Map, Boat and additional statistics are later interface options. A call-and-range-only product requires a separate commercial decision.

Engineering decisions

Agree operating depth, positioning accuracy and coverage, latency, runtime, charge time, actual-scale readability, wrist/cuff coverage and deployment burden. 20 m is a direct-range target. Housing is nominally 54 x 38 x 19.5 mm with a 24 x 24 mm active display. CAD describes envelopes, not validated performance.

Commercial inputs

Current system price, margin, order volume, development budget and launch date are undecided. Evaluate an operator equipment purchase with a possible rental or included-service offer. Account for bands, reference, docks, development, tooling, applicable compliance, logistics, warranty and service. Rev A estimates do not price this system.

Requested partner response

Assess the proposed system for operator-managed boat dives and recommend a feasible first product, the essential trade-offs and a priced scope for the next development stage. Return a product definition, architecture assessment, commercial scope and a proceed/revise/stop recommendation with assumptions and specialist responsibilities.

Concept by Rafael. An aid to buddy procedures, never a safety guarantee. Readings are simulated. No procurement, tooling, manufacturing or physical test program is commissioned.

[Interactive presentation and complete project downloads](#)
Presentation strategy, design & production by The AI Doc